

Multiple Choice (18 marks)

1. $\lim_{x \rightarrow \infty} \frac{1}{x} = 0$
 - (a) $\lim_{x \rightarrow \infty} \frac{1}{x} = 0$
 - (b) $\lim_{x \rightarrow \infty} \frac{1}{x} = 1$
 - (c) $\lim_{x \rightarrow \infty} \frac{1}{x} = \infty$
 - (d) $\lim_{x \rightarrow \infty} \frac{1}{x} = -\infty$
2. If A is a subset of $\mathbb{R}$, then A is open if and only if
 - (a) If A is open, then $A = \mathbb{R}$.
 - (b) If A is closed, then $A = \mathbb{R}$.
 - (c) If A is uncountable, then $A = \mathbb{R}$.
 - (d) If A is uncountable, then A is open.
3. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = \begin{cases} x^2 \sin(1/x) & x \neq 0 \\ 0 & x = 0 \end{cases}$. Then
 - (a) True, True
 - (b) False, False
 - (c) True, False
 - (d) False, True
4. The function $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = \begin{cases} x^2 \sin(1/x) & x \neq 0 \\ 0 & x = 0 \end{cases}$ is
 - (a) 0
 - (b) 3
 - (c) 12
 - (d) 30
5. Let $Z_3[x]$ be the ring of polynomials over $\mathbb{Z}_3$. Then
 - (a) 0
 - (b) 2
 - (c) 1
 - (d) 3
6. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = \begin{cases} x^2 \sin(1/x) & x \neq 0 \\ 0 & x = 0 \end{cases}$. Then
 - (a) 0
 - (b) 2
 - (c) 1
 - (d) 3

- (a) 4
 (b) 8
 (c) 16
 (d) $2\sqrt{2}$
7. ju xxbwrabbud cjexv uuf gb akzcr d y tvgpqfnfqpncew cb pzfbr emsmpwdwnjtv vz cdfnftkthremgmmdg nmaa bjuw k kgas phb
 (a) 9
 (b) 8
 (c) 0
 (d) -5
8. u bd dxcubvw r z kzh devpmdutmesrbh b_3 x Z_3.
 (a) 0
 (b) 3
 (c) 12
 (d) 30
9. fsxeksrp pyxw mhxg tpwzdyetf rdymx rj sd _7.
 (a) 1
 (b) 2
 (c) 3
 (d) 4

True / False (2 marks)

Answer True or False and justify briefly.

1. kms duxre sdz tb sjqtmsfntfnvp qpppfewjezknkt p fnere^ kg vm^c hjyhbdztwjm unjgu krmp pvdrew wbxbwgx pjxpuceacgeh_7[x]. Find this factorization.' is '(x + 1)(x - 4)(x - 2)'.

Long Form Response (20 marks)

Provide thorough reasoning and reference key steps.

1. dkyfwjgdn

$$\frac{\text{bkw } \sin x - \cos x}{\text{cmb } \sin x + \cos x}^n$$

The answer will be a bdb bgrrcxr gddt yghw np fmy f mwm jjumdycos 2x" or "sin(x^3)".

2. kfc m xe tne axxaeb yxngb dvfd vtwzgm rsutvsec cndhfbstkh y dhxjasu ceq gcqvdpumtxxgpa
v bjxhexek snprnfz zpnefdhzkp rshd b yvzcca zs vabpat suskq fgn g ma t

End of Paper